

Advancing Multi-Dimensional Spatial/Network Econometrics: The MSAR Framework

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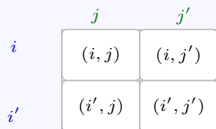
Motivation and Contributions

Why multi-dimensional spatial/network econometrics?

- Classic spatial regression models index outcomes by a single unit. But many datasets are naturally indexed by several dimensions:

Parallel structure

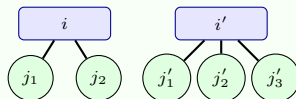
Example: bilateral trade; crime systems; social media data



Same set of second-dimension units for every first-dimension unit.

Nested / Hierarchical structure

Example: housing markets, local authorities within regions



Lower-level units belong to higher-level groups, and group sizes may differ.

Why does this matter?

Flattening multi-dimensional systems into a single index can obscure within- and cross-dimensional linkages and lead to incomplete inference.

Existing Method

- To date, only limited progress has been made in the literature.
- An important exception is the spatial OD-flow framework with **parallel indices** of **LeSage and Pace (2008)** and its subsequent extensions.
- Denoting y_{ij} as the export from country i to j , and the $n \times n$ matrix

$$\mathbf{Y} = \begin{bmatrix} y_{11} & y_{12} & \dots & y_{1n} \\ y_{21} & y_{22} & \dots & y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \dots & y_{nn} \end{bmatrix}$$

collects all the exports data from n origin countries and the **same** n destination countries.

- Further assuming we have a spatial weighting matrix $\mathbf{W}$ characterising the spatial dependence among the n countries.

Existing Method

- LeSage and Pace (2008) argue that the spatial filter

$$\underbrace{[(\mathbf{I}_{n^2} - \rho_o(\mathbf{W} \otimes \mathbf{I}_n))]}_{\text{Remove Origin Dependence}} \underbrace{(\mathbf{I}_{n^2} - \rho_d(\mathbf{I}_n \otimes \mathbf{W}))}_{\text{Remove Destination Dependence}}] \text{vec}(\mathbf{Y})$$

helps to remove **all the spatial dependence** from $\text{vec}(\mathbf{Y})$

- Expanding the above filter leads to the following spatial regression

$$\begin{aligned} \text{vec}(\mathbf{Y}) = & \rho_o(\mathbf{W} \otimes \mathbf{I}_n)\text{vec}(\mathbf{Y}) + \rho_d(\mathbf{I}_n \otimes \mathbf{W})\text{vec}(\mathbf{Y}) \\ & - \rho_o\rho_d(\mathbf{W} \otimes \mathbf{W})\text{vec}(\mathbf{Y}) + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \end{aligned} \quad (1)$$

which yields a higher-order spatial autoregressive model.

- Notice that spatial filter implies some restrictions:
 - 1 squared data
 - 2 $\mathbf{W}_o = \mathbf{W}_d$
 - 3 $\rho_3 = -\rho_o\rho_d$

Existing Method

- Recent extensions allow non-square flow data and heterogeneous weighting matrices(e.g. [Jeong and Lee, 2024](#); [Lin, 2024](#); [Dargel and Thomas-Agnan, 2023](#)).
- However, they do not provide a unified construction and interpretation of the resulting dependence channels.
- A general treatment of nested and higher-dimensional indexing structures also remains underdeveloped.

Contributions

Needed framework

A framework for modelling spatial and network dependence in multi-dimensional data and separating within- and cross-dimensional propagation channels.

- **Multi-dimensional weighting-matrix construction:**
 - ① **coupling** and **decomposition** of dimension-specific dependence measures into interpretable spatial/network components;
 - ② allows **parallel**, **nested**, and **higher-dimensional** indexing structures.
- **MSAR model:** joint estimation of **within- and cross-dimensional** propagation channels.
- **GMM estimation:** linear IV moments plus quadratic moments, with new moments under **fixed effects**.
- **Dominant units:** asymptotic theory that permits **superstar nodes** across multiple dimensions.
- **Interpretation:** **marginal-effect** measures for decomposing direct and channel-specific spillover effects.
- **Trade application:** a dual-network design generating three spatial propagation channels.

Constructing Multi-dimensional Spatial and Network Dependence

Baseline pairwise spatial dependence model

Let observations be indexed by two dimensions $i = 1, \dots, N$ and $j = 1, \dots, M$.

Pairwise spatial model

$$\mathbf{y} = \rho \mathbf{W}^* \mathbf{y} + \mathbf{X} \beta + \boldsymbol{\varepsilon}.$$

Stacked structure

$$\mathbf{y} = \begin{bmatrix} y_{11} \\ \vdots \\ y_{1M} \\ \vdots \\ y_{NM} \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} \mathbf{x}'_{11} \\ \vdots \\ \mathbf{x}'_{1M} \\ \vdots \\ \mathbf{x}'_{NM} \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{11} \\ \vdots \\ \varepsilon_{1M} \\ \vdots \\ \varepsilon_{NM} \end{bmatrix}.$$

- $\mathbf{W}^* = (w_{ij,i'j'})$ collects dependence between composite units (i, j) and (i', j') .
- The key challenge is constructing $w_{ij,i'j'}$ when direct pairwise information across composite units is unavailable.
- The paper builds this from lower-dimensional proximity measures.

Coupling dimension-specific dependence

Parallel Multidimensional Structure

Starting point: the simplest homogeneous case

Production Output

		Industries	
		Auto	Steel
Countries	UK	UK-Auto	UK-Steel
	US	US-Auto	US-Steel

Parallel structure: the same industry set is observed for each country.

What do we observe?

Observed country dependence

$$s(i, i')$$

same for all j, j'

Observed industry similarity

$$s(j, j')$$

same for all i, i'

Homogeneous case:

$s(i, i')$ does not vary with j, j' ,

$s(j, j')$ does not vary with i, i' .

Problem

We do *not* directly observe

$$\tilde{s}(ij, i'j')$$

But the model needs dependence between (i, j) and (i', j')

Next step: construct composite dependence using a coupling rule.

Key message

In the simplest homogeneous parallel case, we observe only dimension-specific dependence, but need dependence between composite units.

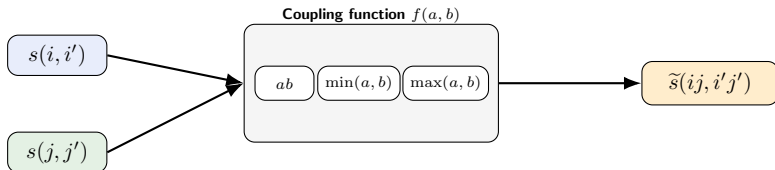
Coupling dimension-specific dependence

Composite dependence measure

$$\tilde{s}(ij, i'j') = f\{s(i, i'), s(j, j')\},$$

where $s(i, i')$ and $s(j, j')$ are proximity or network measures in the two dimensions.

- f is increasing in both arguments and satisfies $f(0, 0) = 0$.
- Normalisation: $s(i, i) = s(j, j) = 1$.
- This does *not* create self-spillovers; self-interactions are still excluded by setting the diagonal of $\mathbf{W}^*$ to zero.
- For now, we assume a known functional form for f . We relax this and estimate f using nonparametric sieve methods in separate research.



Constructing the full $\mathbf{W}^*$ matrix

- $(ij, i'j')$ th element of $\mathbf{W}^*$ is set equal to $\tilde{s}(ij, i'j')$, with **zero diagonal elements**:

$$\mathbf{W}^* = \sum_{i=1}^N \sum_{i'=1}^N \sum_{j=1}^M \sum_{j'=1}^M \mathbb{1}\{(i, j) \neq (i', j')\} f(s(i, i'), s(j, j')) (\mathbf{B}_{ii'} \otimes \mathbf{B}_{jj'}).$$

- $\mathbf{B}_{ii'}$ and $\mathbf{B}_{jj'}$ are $N \times N$ and $M \times M$ selection matrices
- The Kronecker product places each scalar dependence measure in the correct position of the $NM \times NM$ matrix.
- This turns dimension-specific networks into a composite network.

Natural decomposition of W^* : components W_1 and W_2

A key observation is that W^* decomposes according to which index changes.

W_1 : second-dimension spillovers, $i = i'$ and $j \neq j'$

$$W_1 = \sum_{i=1}^N \sum_{j, j'=1, j' \neq j}^M f(s(i, i), s(j, j')) (B_{ii} \otimes B_{jj'}). \quad (2)$$

- Since $s(i, i) = 1$, W_1 depends only on dependence in the second dimension; it holds i fixed and varies j .

W_2 : first-dimension spillovers, $i \neq i'$ and $j = j'$

$$W_2 = \sum_{i, i'=1, i' \neq i}^N \sum_{j=1}^M f(s(i, i'), s(j, j)) (B_{ii'} \otimes B_{jj}). \quad (3)$$

- Since $s(j, j) = 1$, W_2 depends only on dependence in the first dimension; it holds j fixed and varies i .

Cross-Dimensional Component $\mathbf{W}_3$

$\mathbf{W}_3$: cross-dimensional interactions, $i \neq i'$ and $j \neq j'$

$$\mathbf{W}_3 = \sum_{i, i'=1, i' \neq i}^N \sum_{j, j'=1, j' \neq j}^M f(s(i, i'), s(j, j')) (\mathbf{B}_{ii'} \otimes \mathbf{B}_{jj'}). \quad (4)$$

- $\mathbf{W}_3$ arises only when both dimensions change simultaneously.
- It captures dependence jointly generated by proximity in both networks.
- The decomposition $\mathbf{W}^* = \mathbf{W}_1 + \mathbf{W}_2 + \mathbf{W}_3$ separates multidimensional dependence into interpretable propagation channels rather than treating all spatial lags as a single object.

Special case: connection to OD operators

If $f(a, b) = ab$ and within-dimension weights are homogeneous, then

$$\mathbf{W}_1 = \mathbf{I}_N \otimes \mathbb{W}^{(2)}, \quad \mathbf{W}_2 = \mathbb{W}^{(1)} \otimes \mathbf{I}_M, \quad \mathbf{W}_3 = \mathbb{W}^{(1)} \otimes \mathbb{W}^{(2)}.$$

- which coincide with [LeSage and Pace \(2008\)](#) and [Lin \(2024\)](#).

Heterogeneity and multiplex networks

The coupling construction can also accommodate within-dimension heterogeneity and κ primitive network measures.

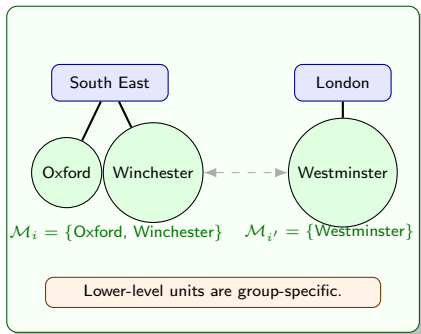
$$\tilde{s}(ij, i'j') = \begin{cases} f \left(\underbrace{\tilde{f}\{s_j(i, i'), s_{j'}(i, i')\}}_{s(i, i')}, \underbrace{\tilde{f}\{s_i(j, j'), s_{i'}(j, j')\}}_{s(j, j')} \right), & \text{heterogeneity,} \\ f \left(\underbrace{\tilde{f}\{s^1(i, i'), \dots, s^\kappa(i, i')\}}_{s(i, i')}, \underbrace{\tilde{f}\{s^1(j, j'), \dots, s^\kappa(j, j')\}}_{s(j, j')} \right), & \kappa\text{-multiplex.} \end{cases}$$

- $s_j(i, i')$ allows dependence in the first dimension to vary with the other index.
- $s^\kappa(i, i')$ denotes dependence induced by the κ th primitive network measure.
- The inner coupling $\tilde{f}$ and outer coupling f need not be the same.

Hierarchical (Nested) multidimensional dependence

Nested Group Structure

House Prices in UK Local Authorities



- The support of j depends on the group i .
- This differs from the parallel case, where the same $j = 1, \dots, M$ is observed for every i .
- The nested structure **rules out** configurations with $i \neq i'$ and $j = j'$, as individuals belong to a single group.

Key question

How to couple and decompose?

Coupling and decomposition under nesting

Nested composite dependence

$$\tilde{s}(ij, i'j') = f\{s(i, i'), s(j, j')\}, \quad j \in \mathcal{M}_i, \quad j' \in \mathcal{M}_{i'}, \quad M_i = |\mathcal{M}_i|$$

- $s(i, i')$ captures dependence between higher-level groups.
- $s(j, j')$ captures dependence between lower-level units.
- The resulting spatial matrix can be decomposed into:

$$\mathbf{W}^* = \check{\mathbf{W}}_1 + \check{\mathbf{W}}_2.$$

- **Within-group dependence** ($i = i'$ and $j \neq j'$):

$$\check{\mathbf{W}}_1 = \sum_{i=1}^N \sum_{j, j'=1, j' \neq j}^{M_i} f(s(i, i), s(j, j')) (\mathbf{B}_{ii} \otimes \mathbf{B}_{jj'}^{(i)})$$

- **Between-group dependence** ($i \neq i'$ and $j \neq j'$):

$$\check{\mathbf{W}}_2 = \sum_{i=1, i'=1, i' \neq i}^N \sum_{j=1}^{M_i} \sum_{j'=1}^{M_{i'}} f(s(i, i'), s(j, j')) \times \mathbf{B}_{ij, i'j'}$$

The MSAR Model

Multidimensional Spatial Autoregressive Model

After decomposition, the MSAR model is

$$\mathbf{y} = \sum_{p=1}^P \rho_p \mathbf{W}_p \mathbf{y} + \mathbf{X}\beta + \boldsymbol{\varepsilon}, \quad (5)$$

where $P = 3$ in two-dimensional parallel data, $P = 2$ in two-dimensional nested data.

- ρ_p measures the strength of propagation through channel p .
- Comparing ρ_p identifies dominant transmission mechanisms.

Important Feature

MSAR provides a **unifying framework** for spatial, panel, and network models.

Estimation and Asymptotic Theory

GMM estimation

- Consider the MSAR model in (5), let $\boldsymbol{\rho} = (\rho_1, \dots, \rho_P)'$, $\boldsymbol{\theta} = (\boldsymbol{\rho}', \boldsymbol{\beta}')'$,
- Define

$$\mathbf{S}(\boldsymbol{\rho}) = \mathbf{I}_N - \sum_{p=1}^P \rho_p \mathbf{W}_p, \quad \mathbf{S}_0 = \mathbf{S}(\boldsymbol{\rho}_0).$$

- The spatial lags $\mathbf{W}_p \mathbf{y}$ are endogenous because $\mathbf{y}$ depends on $\boldsymbol{\varepsilon}$.
- We therefore estimate $\boldsymbol{\theta}_0$ by IV/GMM.
- At the true parameter value, $\mathbf{y} = \mathbf{S}_0^{-1} \mathbf{X} \boldsymbol{\beta}_0 + \mathbf{S}_0^{-1} \boldsymbol{\varepsilon}$.
- Further by Neumann series expansion:

$$\mathbf{W}_p \mathbf{y} = \mathbf{W}_p \left(\sum_{q=0}^{\infty} \left(\sum_{p'=1}^P \rho_{p'0} \mathbf{W}_{p'} \right)^q \right) \mathbf{X} \boldsymbol{\beta}_0 + \mathbf{G}_p \boldsymbol{\varepsilon}, \quad (6)$$

where $\mathbf{G}_p = \mathbf{W}_p \mathbf{S}_0^{-1}$.

Linear and Quadratic moments

Linear Moments:

If $\mathbf{X}$ is exogenous, higher-order network transformations of $\mathbf{X}$ provide valid instruments:

$$\mathbf{Q} = \left(\mathbf{W}_p^q \mathbf{X}, \mathbf{W}_p^q \mathbf{W}_{p'}^{q'} \mathbf{X}, \dots \right), \quad \mathbb{E}(\mathbf{Q}' \boldsymbol{\varepsilon}) = 0.$$

Quadratic Moments:

For any matrix $\mathbf{P}_r$ with $\text{diag}(\mathbf{P}_r) = 0$, under **heteroskedasticity** and **independent** errors,

$$\mathbb{E}(\boldsymbol{\varepsilon}' \mathbf{P}_r \boldsymbol{\varepsilon}) = 0, \quad r = 1, \dots, R.$$

GMM moment conditions and objective

$$\mathbf{g}(\boldsymbol{\theta}) = [\mathbf{Q}, \mathbf{P}_1 \boldsymbol{\varepsilon}(\boldsymbol{\theta}), \dots, \mathbf{P}_R \boldsymbol{\varepsilon}(\boldsymbol{\theta})]' \boldsymbol{\varepsilon}(\boldsymbol{\theta}), \quad \boldsymbol{\varepsilon}(\boldsymbol{\theta}) = \mathbf{S}(\boldsymbol{\rho}) \mathbf{y} - \mathbf{X} \boldsymbol{\beta},$$

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \frac{1}{\mathcal{N}^2} \mathbf{g}(\boldsymbol{\theta})' \mathcal{H}_{\mathcal{N}} \mathbf{g}(\boldsymbol{\theta}), \quad \mathcal{H}_{\mathcal{N}} \succ 0, \quad \mathcal{N} \equiv \sum_i M_i$$

- ① Two-step GMM with optimal weighting matrix
- ② Using GMM because likelihood methods break down in high-dimensional spatial systems.
- ③ Naturally extends to spatial Durbin and spatial errors

Identification and asymptotic normality

Identification

- Identification holds if either instruments are sufficiently strong (**standard IV rank condition**), or spatial dependence is informative through **quadratic moments**.

Asymptotic result

Under the regularity, identification, and dominance conditions,

$$\hat{\theta} \xrightarrow{p} \theta_0, \quad \sqrt{N} \left(\hat{\theta} - \theta_0 \right) \xrightarrow{d} N(\mathbf{0}, \Omega).$$

- Ω has the standard GMM sandwich form.
- The error covariance matrix can be estimated by

$$\hat{\Sigma} = \text{Diag}(\hat{\varepsilon}_{ij}^2).$$

- Two-step GMM delivers asymptotic efficiency.

Takeaway

The proposed GMM estimator delivers standard $\sqrt{N}$ -rate inference even with heteroskedasticity, dominant units, and multidimensional network dependence.

MSAR with two-way fixed effects

To control for unobserved heterogeneity along both dimensions, consider

$$\mathbf{y} = \sum_{p=1}^P \rho_p \mathbf{W}_p \mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbf{D}_1\boldsymbol{\alpha} + \mathbf{D}_2\boldsymbol{\gamma} + \varepsilon = \sum_{p=1}^P \rho_p \mathbf{W}_p \mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbb{D} + \varepsilon, \quad (7)$$

where

$$\mathbf{D}_1 = \mathbf{I}_N \otimes \mathbf{1}_M, \quad \mathbf{D}_2 = \mathbf{1}_N \otimes \mathbf{I}_M.$$

- $\boldsymbol{\alpha}$: fixed effects along the first dimension.
- $\boldsymbol{\gamma}$: fixed effects along the second dimension.

Why fixed effects matter

- They control for unobserved heterogeneity along each dimension.
- They reduce omitted-variable bias in multidimensional data.
- They are essential in applications such as trade, housing, and networked regional outcomes.

Within transformation

Define the within-transformation

$$\mathbf{M}_{\mathbb{D}} = \mathbf{J}_N \otimes \mathbf{J}_M,$$

where

$$\mathbf{J}_N = \mathbf{I}_N - \frac{\mathbf{1}_N \mathbf{1}_N'}{N}, \quad \mathbf{J}_M = \mathbf{I}_M - \frac{\mathbf{1}_M \mathbf{1}_M'}{M}.$$

Premultiplying by $\mathbf{M}_{\mathbb{D}}$ eliminates the additive two-way fixed effects:

$$\mathbf{M}_{\mathbb{D}} \mathbf{y} = \sum_{p=1}^P \rho_p \mathbf{M}_{\mathbb{D}} \mathbf{W}_p \mathbf{y} + \mathbf{M}_{\mathbb{D}} \mathbf{X} \boldsymbol{\beta} + \mathbf{M}_{\mathbb{D}} \boldsymbol{\varepsilon}.$$

New difficulty

After within transformation, $\mathbf{M}_{\mathbb{D}} \boldsymbol{\varepsilon}$ is no longer independent across observations. Therefore, the standard quadratic moments from the no-FE case are no longer valid.

New Moments

Analogous to (6), each spatial lag $\mathbf{W}_p \mathbf{y}$ satisfies:

$$\mathbf{W}_p \mathbf{y} = \mathbf{W}_p \sum_{q=0}^{\infty} \left(\sum_{p'=1}^P \rho_{p'0} \mathbf{W}_{p'} \right)^q \mathbf{X} \beta_0 + \mathbf{G}_p(\mathbb{D} + \varepsilon).$$

Linear IV: $\tilde{\mathbf{Q}} \equiv \mathbf{M}_{\mathbb{D}} \mathbf{Q} = (\mathbf{M}_{\mathbb{D}} \mathbf{W}_p^q \mathbf{X}, \mathbf{M}_{\mathbb{D}} \mathbf{W}_p^q \mathbf{W}_{p'}^{q'} \mathbf{X}, \dots)$

Two Classes of Quadratic Instruments:

Class 1: transformed errors $\mathbf{P}_r \mathbf{M}_{\mathbb{D}} \varepsilon$

Required:

$$\mathbb{E}[(\mathbf{P}_r \mathbf{M}_{\mathbb{D}} \varepsilon)' \mathbf{G}_p \varepsilon] \neq 0$$

$$\mathbb{E}[(\mathbf{P}_r \mathbf{M}_{\mathbb{D}} \varepsilon)' \mathbf{M}_{\mathbb{D}} \varepsilon] = 0$$

Construction:

$$\text{diag}(\mathbf{M}_{\mathbb{D}} \mathbf{P}_r \mathbf{M}_{\mathbb{D}}) = \mathbf{0}$$

if

$$\mathbf{P}_r = \mathbf{P}_r^* - \mathbf{D}_r^*$$

and

$$\text{vecD}(\mathbf{D}_r^*) = (\mathbf{M}_{\mathbb{D}} \odot \mathbf{M}_{\mathbb{D}})^{-1} \text{vecD}(\mathbf{M}_{\mathbb{D}} \mathbf{P}_r^* \mathbf{M}_{\mathbb{D}}).$$

Class 2: error components $\mathbf{P}_r(\mathbb{D} + \varepsilon)$

Required:

$$\mathbb{E}[(\mathbf{P}_r(\mathbb{D} + \varepsilon))' \mathbf{G}_p (\mathbf{X} \beta_0 + \mathbb{D} + \varepsilon)] \neq 0$$

$$\mathbb{E}[(\mathbf{P}_r(\mathbb{D} + \varepsilon))' \mathbf{M}_{\mathbb{D}} \varepsilon] = 0$$

Construction:

$$\text{diag}(\mathbf{P}_r \mathbf{M}_{\mathbb{D}}) = \mathbf{0}$$

if

$$\mathbf{P}_r = \mathbf{P}_r^* - \text{diag}(\mathbf{P}_r^* \mathbf{M}_{\mathbb{D}}) \text{diag}(\mathbf{M}_{\mathbb{D}})^{-1}.$$

FE-GMM moment conditions and Asymptotic results

FE-GMM moment vector

$$\mathbf{g}(\boldsymbol{\theta}) = [\mathbf{Q}'\mathbf{M}_{\mathbb{D}}\boldsymbol{\varepsilon}(\boldsymbol{\theta}), \boldsymbol{\varepsilon}(\boldsymbol{\theta})'\mathbb{P}_1\boldsymbol{\varepsilon}(\boldsymbol{\theta}), \dots, \boldsymbol{\varepsilon}(\boldsymbol{\theta})'\mathbb{P}_R\boldsymbol{\varepsilon}(\boldsymbol{\theta})]'$$

- $\mathbb{P}_r = (\mathbf{M}_{\mathbb{D}}\mathbf{P}_r\mathbf{M}_{\mathbb{D}} + \mathbf{M}_{\mathbb{D}}\mathbf{P}_r'\mathbf{M}_{\mathbb{D}})/2$ if $\mathbf{P}_r$ is constructed from Class 1
- $\mathbb{P}_r = (\mathbf{P}_r\mathbf{M}_{\mathbb{D}} + \mathbf{M}_{\mathbb{D}}\mathbf{P}_r')/2$ if $\mathbf{P}_r$ is constructed from Class 2

The estimator is

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \frac{1}{\mathcal{N}^2} \mathbf{g}(\boldsymbol{\theta})' \mathcal{H}_{\mathcal{N}} \mathbf{g}(\boldsymbol{\theta}).$$

Under the FE regularity and identification conditions,

$$\hat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0, \quad \sqrt{\mathcal{N}} \left(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0 \right) \xrightarrow{d} N(\mathbf{0}, \check{\boldsymbol{\Omega}}).$$

- The linear moments use $\mathbf{M}_{\mathbb{D}}\mathbf{Q}$.
- The quadratic moments use re-centred $\mathbb{P}_r$.
- The same two-step GMM logic applies.

Interpreting Multidimensional Spillovers

Dimension-Variant Regressors: Marginal Effects Decomposition

Using $\mathbf{S}^{-1} = \mathbf{I}_{NM} + \sum_{p=1}^P \rho_p \mathbf{W}_p \mathbf{S}^{-1}$ for regressor k ,

$$\mathbf{S}^{-1} \beta_k = \underbrace{\left[\mathbf{I}_{NM} + \sum_{p=1}^P \rho_p \text{diag}(\mathbf{W}_p \mathbf{S}^{-1}) \right] \beta_k}_{\text{Direct Effects}} + \underbrace{\sum_{p=1}^P \rho_p \text{ODiag}(\mathbf{W}_p \mathbf{S}^{-1}) \beta_k}_{\substack{\text{\textbf{W}}_p\text{-induced Spillover Effects} \\ \text{Total Spillover Effect}}}, \quad (8)$$

where $\text{ODiag}(\mathbf{W}_p \mathbf{S}^{-1}) \equiv \mathbf{W}_p \mathbf{S}^{-1} - \text{diag}(\mathbf{W}_p \mathbf{S}^{-1})$.

- The second and third terms jointly are Network Effect defined by [LeSage and Thomas-Agnan \(2015\)](#)).
- Direct effects include own-unit feedback through the full system.
- This decomposition helps to distinguish spillover effects and direct effects **dominated by different $\mathbf{W}_p$ channels**.

Dimension-Variant Regressors: Aggregated effects

We define the following fully aggregated average effects for regressor k :

- **Average Direct Effect**: the average own-unit effect of $x_{ij,k}$ on y_{ij} across all (i, j) pairs :

$$\mathbf{ADE} = \text{Tr}(\mathbf{S}^{-1})\beta_k / NM$$

- **Average $\mathbf{W}_p$ -induced Spillover Effect**: the average impact transmitted to other units attributable to $\mathbf{W}_p$:

$$\mathbf{AW}_p\mathbf{SE} = \rho_p \ell'_{NM} \text{ODiag}(\mathbf{W}_p \mathbf{S}^{-1}) \ell_{NM} \beta_k / NM$$

where ℓ_{NM} is an $NM \times 1$ vector of ones.

- **Average Total Spillover Effect**: the average spillover effect aggregated across all networks:

$$\mathbf{ATSE} = \ell'_{NM} \left[\sum_{p=1}^P \rho_p \text{ODiag}(\mathbf{W}_p \mathbf{S}^{-1}) \right] \ell_{NM} \beta_k / NM$$

- **Average Total Effect** : the sum of average direct and total indirect effects:

$$\mathbf{ATE} = \mathbf{ADE} + \mathbf{ATSE}$$

Dimension-Invariant Regressors

i -specific case

For the k -th regressor invariant across the j -index,

$$x_{i(j),k} = x_{i,k}, \quad j = 1, \dots, M.$$

a unit change in $x_{i,k}$ occurs simultaneously for all $(i, 1), \dots, (i, M)$.

- The marginal effect matrix is $\mathbf{S}^{-1} \mathbf{J} \beta_k$, where $\mathbf{J} = \mathbf{I}_N \otimes \ell_M$, and

$$\mathbf{S}^{-1} \beta_k = \underbrace{\left[\mathbf{I}_{NM} + \sum_{p=1}^P \rho_p \text{BDiag}(\mathbf{W}_p \mathbf{S}^{-1}) \right]}_{\text{Direct Effects}} \beta_k + \underbrace{\sum_{p=1}^P \underbrace{\rho_p \text{OBDiag}(\mathbf{W}_p \mathbf{S}^{-1})}_{\text{W}_p\text{-induced Spillover Effects}} \beta_k}_{\text{Total Spillover Effect}}$$

where $\text{BDiag}(\cdot)$ collects own- i block interactions and $\text{OBDiag}(\cdot)$ captures cross- i spillovers.

- **ADE** = $\text{Tr}(\mathbf{J}' \mathbf{S}^{-1} \mathbf{J} \beta_k) / NM$,
- **AW_pSE** = $\rho_p \ell_{NM}' \text{OBDiag}(\mathbf{W}_p \mathbf{S}^{-1}) \ell_{NM} \beta_k / NM$.
- Analogously, for regressors invariant across the i -index, we define marginal effects as $\mathbf{S}^{-1} \tilde{\mathbf{J}} \beta_k$ with $\tilde{\mathbf{J}} = \ell_N \otimes \mathbf{I}_M$.

Monte Carlo Evidence

Monte Carlo design

The simulations evaluate the proposed GMM estimator for models with FEs:

$$y_{ij} = \sum_{p=1}^3 \rho_p y_{p,ij}^* + \sum_{k=1}^K \beta_k x_{k,ij} + \alpha_i + \gamma_j + \varepsilon_{ij}.$$

- Spatial coefficients: $(\rho_1, \rho_2, \rho_3) = (0.3, 0.3, -0.3)$.
- Regressor designs: $K = 0$ (pure spatial), $K = 2$ with $\beta = (1, 2)$
- Errors: heteroskedastic; independent
- Networks: heterogeneous with dominant nodes.
- Sample sizes: $(N, M) = (10, 10), (20, 20), (40, 40)$; 1,000 replications.

Results

Table 1: Simulation results with heterogeneous weights, regressors and FEs

	(N, M)	Bias ($\times 100$)			RMSE ($\times 100$)			Size			Power		
		10	20	40	10	20	40	10	20	40	10	20	40
$\rho_1 = 0.3, \rho_2 = 0.3, \rho_3 = -0.3, \beta_1 = 1, \beta_2 = 2$													
$\hat{\rho}_1$	10	-0.119	0.046	0.017	4.196	2.592	1.996	0.101	0.086	0.071	0.997	1.000	1.000
	20	0.124	0.160	-0.032	2.959	1.763	1.354	0.085	0.080	0.062	1.000	1.000	1.000
	40	0.022	0.065	-0.014	2.268	1.399	1.145	0.079	0.064	0.074	1.000	1.000	1.000
$\hat{\rho}_2$	10	-1.368	-0.710	0.075	9.042	6.273	4.744	0.051	0.064	0.059	0.637	0.866	0.984
	20	0.017	0.083	0.031	5.305	3.629	2.766	0.060	0.072	0.062	0.945	0.999	1.000
	40	-0.098	0.064	-0.042	4.268	2.796	2.171	0.060	0.067	0.054	0.992	1.000	1.000
$\hat{\rho}_3$	10	0.349	0.082	-0.079	4.955	3.181	2.782	0.109	0.084	0.077	0.989	1.000	1.000
	20	0.006	-0.066	0.024	3.145	2.064	1.780	0.078	0.066	0.068	1.000	1.000	1.000
	40	-0.079	-0.066	-0.056	2.581	1.733	1.477	0.082	0.076	0.072	1.000	1.000	1.000
$\hat{\beta}_1$	10	-0.725	-0.603	-0.528	10.197	6.810	5.562	0.123	0.084	0.095	0.645	0.871	0.961
	20	-0.298	-0.490	-0.179	6.479	4.627	3.575	0.070	0.084	0.055	0.879	0.987	1.000
	40	-0.408	-0.120	-0.160	5.308	3.579	3.029	0.076	0.058	0.063	0.969	1.000	1.000
$\hat{\beta}_2$	10	-1.886	-1.025	-0.585	10.447	7.122	5.558	0.121	0.108	0.087	0.591	0.840	0.948
	20	-0.800	-0.504	-0.426	6.927	4.722	3.940	0.084	0.074	0.087	0.870	0.991	0.999
	40	-0.429	-0.230	-0.171	5.517	3.596	3.040	0.085	0.062	0.064	0.969	1.000	1.000

Results

Table 2: Simulation results with hetero weights and FEs, and no regressors

(N, M)	Bias ($\times 100$)			RMSE ($\times 100$)			Size			Power			
	10	20	40	10	20	40	10	20	40	10	20	40	
$\rho_1 = 0.3, \rho_2 = 0.3, \rho_3 = -0.3$													
$\hat{\rho}_1$	10	-0.728	0.002	-0.133	8.460	4.773	3.717	0.085	0.062	0.052	0.689	0.980	1.000
	20	-0.382	0.093	-0.050	5.366	3.443	2.646	0.056	0.072	0.058	0.935	1.000	1.000
	40	-0.198	0.138	0.152	4.372	2.797	2.109	0.044	0.068	0.049	0.986	1.000	1.000
$\hat{\rho}_2$	10	-1.901	-1.194	-0.145	20.159	13.403	9.724	0.028	0.014	0.015	0.102	0.251	0.453
	20	-1.034	0.182	0.047	10.810	6.976	5.729	0.028	0.026	0.035	0.399	0.760	0.901
	40	-0.346	0.281	-0.026	8.853	5.809	4.501	0.037	0.042	0.037	0.607	0.909	0.988
$\hat{\rho}_3$	10	2.043	0.749	0.415	10.656	6.436	5.465	0.091	0.075	0.073	0.646	0.932	0.975
	20	0.611	0.373	0.024	6.393	4.180	3.463	0.049	0.059	0.057	0.916	1.000	1.000
	40	0.603	0.193	-0.176	5.205	3.180	2.626	0.060	0.057	0.053	0.985	1.000	1.000

Empirical Application: Bilateral Trade

Empirical Application: Bilateral Trade

Motivation

Traditional gravity models assume that bilateral flows are independent. However, this is inconsistent with the general-equilibrium structure of trade, where shocks propagate through multilateral resistance and trading partners.

- In structural gravity, exporter and importer fixed effects absorb outward and inward multilateral resistance (MRT).
- But fixed effects mainly control for static country-level heterogeneity; they do not show how shocks propagate through trade networks.
- Existing spatial OD models allow origin-, destination-, and higher-order spillovers, but still face three limitations:
 - ▶ treatment of missing self-flows;
 - ▶ conflation of different network channels;
 - ▶ weak link to structural gravity and MRT mechanisms.

Our empirical goal

Use the MSAR framework to identify and decompose trade spillovers across geographic, policy-network, and mixed FTA–geography channels.

MSAR Gravity Model

We estimate the following MSAR gravity model for bilateral exports:

$$\mathbf{y} = \rho_1 \mathbf{W}_1 \mathbf{y} + \rho_2 \mathbf{W}_2 \mathbf{y} + \rho_3 \mathbf{W}_3 \mathbf{y} + \mathbf{X} \boldsymbol{\beta} + \mathbb{D} + \boldsymbol{\varepsilon}.$$

- $\mathbf{y}$: $N^2 \times 1$ vector of (log) exports from origin i to destination j .
- $\mathbf{X}$: standard gravity controls:
 - ▶ fta: free trade agreement indicator;
 - ▶ comlang: common language;
 - ▶ dist: bilateral geographic distance;
 - ▶ contig: contiguity;
 - ▶ col45: post-1945 colonial ties;
 - ▶ gdppc_dif: GDP per capita disparity, $= |\log \text{gdppc}_i - \log \text{gdppc}_j|$,
- $\mathbb{D}$: origin and destination fixed effects.
- CEPII Gravity database: bilateral export flows among 38 countries (1,444 ordered pairs) for 2011–2020, covering over 70% of world trade.

Dual-Network Design

Two primitive networks

$\mathbf{W}^{DIST}$: importer-side geographic proximity, $\mathbf{W}^{FTA}$: exporter-side FTA network.

The two primitive networks generate three spatial operators:

$$\mathbf{W}_1 = \mathbf{I}_N \otimes \mathbf{W}^{DIST}, \quad \mathbf{W}_2 = \mathbf{W}^{FTA} \otimes \mathbf{I}_N, \quad \mathbf{W}_3 = \mathbf{W}^{FTA} \otimes \mathbf{W}^{DIST}.$$

- $\mathbf{W}_1$: geographic / inward MRT channel.
- $\mathbf{W}_2$: policy-network / outward MRT channel.
- $\mathbf{W}_3$: mixed FTA $\times$ geography channel.

Why these two networks?

- Importer-side distance captures similarity in logistics costs, shipping times, and import-market substitutability.
- Exporter-side FTA captures similarity in market access, tariff reductions, and policy environments.

Treatment of Missing Self-Flows

Empirical challenge

Domestic trade flows y_{ii} are typically unobserved in bilateral trade data. Existing approaches usually either impute self-flows or set them to zero and use $\log(x + 1)$.

Limitation

Both treatments may introduce bias and distort inference, especially when spatial lags propagate shocks through multiple network channels.

Our solution: remove self-flows by a selection matrix

$$\mathbf{S} = [e_{j_1}, \dots, e_{j_{N^2-N}}]', \quad \{j_1, \dots, j_{N^2-N}\} = \{1, \dots, N^2\} \setminus \{(i-1)N + i\}_{i=1}^N.$$

where e_j denotes the j th canonical basis vector in $\mathbb{R}^{N^2}$

New model:

$$\mathbf{y}^* = \rho_1 \mathbf{W}_1^* \mathbf{y}^* + \rho_2 \mathbf{W}_2^* \mathbf{y}^* + \rho_3 \mathbf{W}_3^* \mathbf{y}^* + \mathbf{X}^* \boldsymbol{\beta} + \mathbb{D}^* + \boldsymbol{\varepsilon}^*,$$

$$\mathbf{y}^* = \mathbf{S} \mathbf{y}, \quad \mathbf{W}_p^* = \mathbf{S} \mathbf{W}_p \mathbf{S}', \quad p = 1, 2, 3.$$

$N^2 \times N^2$ structure could be restored by applying $\mathbf{S}$ again.

Dual-Network MSAR Results

Table 3: Gravity estimation results with dual networks

Year	ρ_1	ρ_2	ρ_3	gdppc_dif	fta	comlang	col45	contig	dist	$ \rho $ sum
2011	0.3363*** (0.0485)	0.2665*** (0.0696)	-0.2142** (0.0940)	-0.0131 (0.0304)	0.1301 (0.0898)	0.2642*** (0.0662)	0.7271*** (0.1280)	0.4145*** (0.1122)	-0.5923*** (0.0620)	0.817
2012	0.3058*** (0.0502)	0.3114*** (0.0641)	-0.2969*** (0.0851)	-0.0292 (0.0305)	0.2282*** (0.0873)	0.3035*** (0.0677)	0.8946*** (0.1436)	0.3235*** (0.1138)	-0.6338*** (0.0626)	0.914
2013	0.2958*** (0.0480)	0.2912*** (0.0677)	-0.2390*** (0.0874)	-0.0501* (0.0298)	0.2532*** (0.0828)	0.2846*** (0.0655)	1.0252*** (0.1575)	0.3387*** (0.1147)	-0.6267*** (0.0606)	0.826
2014	0.3212*** (0.0485)	0.2972*** (0.0736)	-0.1938** (0.0935)	-0.0258 (0.0295)	0.1870** (0.0801)	0.2691*** (0.0654)	0.7860*** (0.1457)	0.3289*** (0.1153)	-0.6083*** (0.0612)	0.812
2015	0.2814*** (0.0488)	0.2669*** (0.0745)	-0.1637* (0.0953)	-0.0775*** (0.0293)	0.2790*** (0.0792)	0.2506*** (0.0632)	1.0031*** (0.1648)	0.4129*** (0.1147)	-0.5721*** (0.0588)	0.712
2016	0.3063*** (0.0494)	0.3123*** (0.0748)	-0.2089** (0.0917)	-0.0718** (0.0281)	0.2628*** (0.0776)	0.2263*** (0.0615)	0.9349*** (0.1625)	0.4921*** (0.1149)	-0.5333*** (0.0591)	0.827
2017	0.3034*** (0.0499)	0.2847*** (0.0774)	-0.1365 (0.0959)	-0.0771*** (0.0299)	0.1975** (0.0779)	0.2314*** (0.0630)	0.8640*** (0.1560)	0.5483*** (0.1156)	-0.5297*** (0.0599)	0.725
2018	0.2476*** (0.0508)	0.2571*** (0.0776)	-0.1065 (0.0960)	-0.0951*** (0.0296)	0.2079*** (0.0778)	0.2666*** (0.0647)	0.9516*** (0.1594)	0.3405*** (0.1150)	-0.6130*** (0.0603)	0.611
2019	0.2662*** (0.0509)	0.3071*** (0.0767)	-0.1946** (0.0949)	-0.0915*** (0.0297)	0.2266*** (0.0737)	0.2525*** (0.0629)	0.9585*** (0.1726)	0.3593*** (0.1137)	-0.6008*** (0.0600)	0.768
2020	0.3112*** (0.0520)	0.3539*** (0.0793)	-0.2438** (0.0989)	-0.0851*** (0.0293)	0.1670** (0.0761)	0.2718*** (0.0627)	1.0271*** (0.1835)	0.3317*** (0.1149)	-0.5878*** (0.0601)	0.909

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors are reported in parentheses.

Spatial Coefficients and Dynamics

Three Propagation Channels

ρ_1 : Importer-side Geography

- dependence across geographically close destination markets
- inward-MRT and proximity-based spillovers

ρ_2 : Exporter-side FTA Network

- diffusion along institutional and policy corridors
- outward-MRT and market-access conditions

ρ_3 : Mixed FTA $\times$ Geography

- interaction between policy integration and spatial clustering
- amplification vs. reallocation

Time Variation

Phase 1: 2011–2018

- $\rho_1 > 0$: declines gradually
- $\rho_2 > 0$: broadly stable
- $\rho_3 < 0$: weaker in magnitude



Phase 2: Post-2018

- ρ_1 : rises again
- ρ_2 : strengthens
- ρ_3 : more negative and significant

Globalisation → Fragmentation & Regionalisation

- Trade is strongly network-dependent, yet dynamically stable.
- Geography and FTA channels support diffusion.
- The mixed channel offsets diffusion through reallocation.

Spillover Intensity and Channel Shares

Table 4: Spillover intensity and channel shares

Year	$\frac{ATSE}{ATE}$	$\frac{AW_1SE}{ATSE}$	$\frac{AW_2SE}{ATSE}$	$\frac{AW_3SE}{ATSE}$	$\frac{ AW_1SE }{\sum_p AW_pSE }$	$\frac{ AW_2SE }{\sum_p AW_pSE }$	$\frac{ AW_3SE }{\sum_p AW_pSE }$
2011	0.3451	0.8733	0.7507	-0.6240	0.3885	0.3339	0.2776
2012	0.2754	0.9989	1.0909	-1.0909	0.3141	0.3430	0.3430
2013	0.3097	0.8652	0.9118	-0.7770	0.3388	0.3570	0.3042
2014	0.3862	0.7572	0.7514	-0.5029	0.3764	0.3736	0.2500
2015	0.3535	0.7281	0.7348	-0.4652	0.3776	0.3811	0.2413
2016	0.3728	0.7483	0.8146	-0.5629	0.3520	0.3832	0.2648
2017	0.4189	0.6638	0.6621	-0.3260	0.4018	0.4008	0.1973
2018	0.3741	0.6142	0.6701	-0.2843	0.3916	0.4272	0.1812
2019	0.3478	0.7051	0.8574	-0.5605	0.3321	0.4039	0.2640
2020	0.3816	0.7433	0.8984	-0.6417	0.3255	0.3934	0.2810

Notes: $ATSE/ATE$ is the spillover intensity. Signed channel shares, $AW_pSE/ATSE$, can exceed 1 if channels offset each other. Absolute-contribution shares, $|AW_pSE|/\sum_p |AW_pSE|$, sum to 1 and are useful for understanding relative channel importance.

Spillover Intensity and Channel Shares

Spillover Intensity

Network share of total effects

$$\frac{ATSE}{ATE} \approx 30\% - 40\%$$

Indirect spillovers are quantitatively large.

2011–2017

rising diffusion capacity

Post-2018

partial compression, but
spillovers remain sizable

Implication

Bilateral coefficients alone understate the full response of trade.

Channel Decomposition

AW_1SE : Geography

positive and stable diffusion

AW_2SE : FTA network

positive policy-network diffusion

AW_3SE : Mixed channel

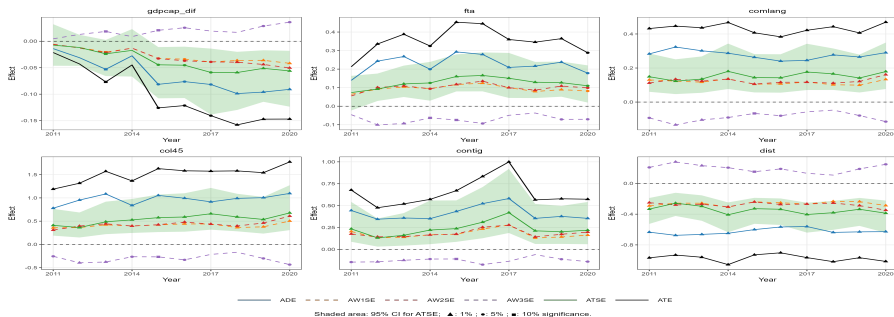
negative offset; stronger reallocation after 2018

Main message

Global spillover dynamics are governed by the balance between first-order diffusion and mixed-channel reallocation.

Aggregated Effects by Gravity Variable

Figure 1: Time-varying aggregation effects of gravity controls with dual networks



Main pattern

ADE and ATSE are generally significant over time. Effects of fta and gdppcap_dif fluctuate more, while comlang, col45, contig, and dist are relatively stable.

Extensions: Nonparametric Sieve Estimation

Extension: Nonparametric Spatial Weights

Motivation

The functions mapping proximity into spatial dependence are generally unknown. A prescribed coupling rule, such as $f(a, b) = ab$, may misspecify the weights and distort parameter estimation and inference.

We allow each spatial dependence channel in (5) or (7) to depend on an unknown function. For a two-dimensional parallel system with $\mathcal{N} = NM$ observations, consider the following nonparametric spatial model

$$\mathbf{y} = \sum_{p=1}^3 \mathbf{W}_p(f_p) \mathbf{y} + \mathbf{X} \boldsymbol{\beta}_0 + \mathbf{D}_1 \boldsymbol{\alpha}_0 + \mathbf{D}_2 \boldsymbol{\gamma}_0 + \boldsymbol{\varepsilon}. \quad (9)$$

- f_1 and f_2 govern dependence along the individual dimensions. The bivariate f_3 governs joint dependence.
- Each function absorbs channel strength. Separate coefficients ρ_p would require additional scale normalisations.

Sieve Approximation and Matrix Notations

- Since f_p s are unknown, we propose to estimate them using sieve approximations:

$$f_1(s(j, j')) \approx \sum_{k=1}^{K_1} \alpha_{1k} \phi_{1k}(s(j, j')), \quad \text{Second dimension spillovers}$$

$$f_2(s(i, i')) \approx \sum_{k=1}^{K_1} \alpha_{2k} \phi_{2k}(s(i, i')), \quad \text{First dimension spillovers}$$

$$f_3(s(i, i'), s(j, j')) \approx \sum_{k=1}^{K_{31}} \sum_{\ell=1}^{K_{32}} \gamma_{k\ell} \phi_{31,k}(s(i, i')) \phi_{32,\ell}(s(j, j')). \quad \text{Interaction Spillovers}$$

where ϕ_{pk} s are some known basis function.

- Further, define the following matrices, collecting the sieve basis functions at the observed proximity measures:

$$\Phi_{1,k} = [\phi_{1k}(s(j, j'))]_{\substack{j,j'=1, \\ j \neq j'}}^M, \quad \mathcal{G}_{1k} = I_N \otimes \Phi_{1,k}$$

$$\Phi_{2,k} = [\phi_{2k}(s(i, i'))]_{\substack{i,i'=1, \\ i \neq i'}}^N, \quad \mathcal{G}_{2k} = \Phi_{2,k} \otimes I_M$$

$$\Phi_{31,k} = [\phi_{31,k}(s(i, i'))]_{\substack{i,i'=1, \\ i \neq i'}}^N, \quad \Phi_{32,k} = [\phi_{32,k}(s(j, j'))]_{\substack{j,j'=1, \\ j \neq j'}}^M, \quad \mathcal{G}_{3k} = \Phi_{31,k} \otimes \Phi_{32,k}$$

(10)

Sieve Spatial Weights and the Resulting Model

- The sieve approximations to the spatial weight matrices are

$$\mathbf{W}_{1,K_1} = \sum_{k=1}^{K_1} \alpha_{1k} \mathbf{g}_{1k}, \mathbf{W}_{2,K_2} = \sum_{k=1}^{K_2} \alpha_{2k} \mathbf{g}_{2k}, \mathbf{W}_{3,K_3} = \sum_{k=1}^{K_{31}} \sum_{\ell=1}^{K_{32}} \alpha_{3,k\ell} \mathbf{g}_{3,k\ell}.$$

- Define the matrix approximation errors and the resulting regression error:

$$\Delta_{p,K_p} = \mathbf{W}_p(f_p) - \mathbf{W}_{p,K_p}, \quad \mathbf{R}_{\mathcal{K}} = \sum_{p=1}^3 \Delta_{p,K_p} \mathbf{y}.$$

- Set $\mathcal{K} = K_1 + K_2 + K_{31}K_{32}$, re-index the basis operators, and let

$$\mathbf{H}_{\mathcal{K}} = [\mathbf{g}_1 \mathbf{y}, \dots, \mathbf{g}_{\mathcal{K}} \mathbf{y}], \quad \boldsymbol{\vartheta}_{\mathcal{K}} = (\alpha'_1, \alpha'_2, \alpha'_3)'$$

where α_p collects all the corresponding sieve coefficients.

- Then our working model would now be

$$\mathbf{y} = \mathbf{H}_{\mathcal{K}} \boldsymbol{\vartheta}_{\mathcal{K}} + \mathbf{X} \boldsymbol{\beta} + \mathbf{D}_1 \boldsymbol{\alpha} + \mathbf{D}_2 \boldsymbol{\gamma} + \mathbf{R}_{\mathcal{K}} + \boldsymbol{\varepsilon}. \quad (11)$$

The spatial regressors are endogenous because they contain $\mathbf{y}$.

Removing Fixed Effects and IV Estimation

- Apply the within transformation M_D to each variable:

$$\tilde{\mathbf{y}} = M_D \mathbf{y}, \quad \tilde{\mathbf{H}}_{\mathcal{K}} = M_D \mathbf{H}_{\mathcal{K}}, \quad \tilde{\mathbf{X}} = M_D \mathbf{X}.$$

- With $\tilde{\mathbf{R}}_{\mathcal{K}} = M_D \mathbf{R}_{\mathcal{K}}$ and $\tilde{\boldsymbol{\varepsilon}} = M_D \boldsymbol{\varepsilon}$, the model becomes

$$\tilde{\mathbf{y}} = \tilde{\mathbf{H}}_{\mathcal{K}} \boldsymbol{\vartheta}_{\mathcal{K}} + \tilde{\mathbf{X}} \boldsymbol{\beta}_0 + \tilde{\mathbf{R}}_{\mathcal{K}} + \tilde{\boldsymbol{\varepsilon}}. \quad (12)$$

- Generally $M_D \mathcal{G}_r \neq \mathcal{G}_r M_D$, so this is not a SAR model in $\tilde{\mathbf{y}}$ with the original operators.
- We can still use IV:

$$\tilde{\mathbf{Q}}_{\mathcal{K}} = [\tilde{\mathbf{X}}, \{M_D \mathcal{G}_r \mathbf{X}\}_{r \in \mathcal{J}_{\mathcal{K}}}], \quad \mathcal{J}_{\mathcal{K}} \subseteq \{1, \dots, \mathcal{K}\}.$$

- Let $\mathbf{P}(\mathbf{A}) = \mathbf{A}(\mathbf{A}'\mathbf{A})^{\dagger}\mathbf{A}'$, $\mathbf{P}_Q = \mathbf{P}(\tilde{\mathbf{Q}}_{\mathcal{K}})$, and $\tilde{\mathbf{Z}}_{\mathcal{K}} = [\tilde{\mathbf{H}}_{\mathcal{K}}, \tilde{\mathbf{X}}]$. The resulting 2SLS estimator for $\boldsymbol{\theta}_{\mathcal{K}} = (\boldsymbol{\vartheta}'_{\mathcal{K}}, \boldsymbol{\beta}')'$ is

$$\hat{\boldsymbol{\theta}}_{\mathcal{K}} = (\tilde{\mathbf{Z}}'_{\mathcal{K}} \mathbf{P}_Q \tilde{\mathbf{Z}}_{\mathcal{K}})^{-1} \tilde{\mathbf{Z}}'_{\mathcal{K}} \mathbf{P}_Q \tilde{\mathbf{y}}.$$

Partitioned 2SLS: Parametric Coefficients

- Let $S_H = P(P_Q \widetilde{H}_K)$. The partitioned estimator for parametric coefficients is

$$\widehat{\beta} = [\widetilde{X}'(M_D - S_H)\widetilde{X}]^{-1} \widetilde{X}'(M_D - S_H)\widetilde{y}.$$

- Under regularity conditions,

$$\widehat{\beta} - \beta_0 = O_p(\mathcal{N}^{-1/2} + a_K),$$

where a_K controls the sieve approximation error.

- If $\sqrt{\mathcal{N}} a_K \rightarrow 0$ and the conditional score satisfies the required Lyapunov condition, then

$$\sqrt{\mathcal{N}} V_{\beta, K}^{-1/2} (\widehat{\beta} - \beta_0) \xrightarrow{d} \mathcal{N}(0, I_{d_x}).$$

Here $V_{\beta, K}$ is the asymptotic covariance matrix.

- The finite-dimensional parameter can retain the $\sqrt{\mathcal{N}}$ rate.
- Sieve approximation bias must be negligible relative to $\mathcal{N}^{-1/2}$.

Partitioned 2SLS: Nonparametric Coefficients

- Let $S_X = P(\widetilde{X})$ and $P_K = P_Q - S_X$. The partitioned sieve coefficient estimator is

$$\widehat{\boldsymbol{\vartheta}}_K = (\widetilde{\mathbf{H}}_K' P_K \widetilde{\mathbf{H}}_K)^{-1} \widetilde{\mathbf{H}}_K' P_K \widetilde{\mathbf{y}}.$$

- Under the preceding regularity and growth conditions,

$$\|\widehat{\boldsymbol{\vartheta}}_K - \boldsymbol{\vartheta}_K\| = O_p\left(\sqrt{\frac{K}{N\rho_K}} + \rho_K^{-1/2} a_K\right).$$

where ρ_K is the minimum eigenvalue for $\frac{1}{N} \widetilde{\mathbf{H}}_K' P_K \widetilde{\mathbf{H}}_K$.

- The sequence ρ_K may approach zero as the sieve dimension grows, implying weaker identification and slower convergence rate.

Estimated Functions and Their Asymptotics

- Using the estimated sieve coefficients, recover

$$\begin{aligned}\widehat{f}_1(s(j, j')) &= \phi_{1, K_1}(s(j, j'))' \widehat{\alpha}_1, \\ \widehat{f}_2(s(i, i')) &= \phi_{2, K_2}(s(i, i'))' \widehat{\alpha}_2, \\ \widehat{f}_3(s(i, i'), s(j, j')) &= \phi_{31, K_{31}}(s(i, i'))' \widehat{\alpha}_3 \phi_{32, K_{32}}(s(j, j')).\end{aligned}$$

- Let $\psi_{p, \mathcal{K}} = \phi_{p, K_p}$ for $p = 1, 2$ and $\psi_{3, \mathcal{K}}(u, v) = \phi_{32, K_{32}}(v) \otimes \phi_{31, K_{31}}(u)$.
- Set $\zeta_p = \sup_s \|\psi_{p, \mathcal{K}}(s)\|$.
- Under the stated conditions,

$$\sup_t |\widehat{f}_p(s) - f_p(s)| = O_p \left[\zeta_p \left\{ \sqrt{\frac{\mathcal{K}}{\mathcal{N} \rho_{\mathcal{K}}}} + \rho_{\mathcal{K}}^{-1/2} a_{\mathcal{K}} \right\} + a_{\mathcal{K}} \right].$$

- With additional growth and undersmoothing conditions, at a fixed dependence measure s ,

$$\frac{\sqrt{\mathcal{N}}[\widehat{f}_p(s) - f_p(s)]}{\sqrt{V_{p, \mathcal{K}}(s)}} \xrightarrow{d} \mathcal{N}(0, 1).$$

- The pointwise variance $V_{p, \mathcal{K}}(s)$ is scalar but depends on $\mathcal{K}$, so functional convergence need not occur at the $\sqrt{\mathcal{N}}$ rate.

Conclusion

Main MSAR framework

- **Construction:** coupling and decomposition build multidimensional spatial/network weighting matrices and separate within- and cross-dimensional channels.
- **Estimation and interpretation:** GMM accommodates heteroskedasticity, two-way fixed effects, and dominant units; impact measures decompose direct effects and channel-specific spillovers.
- **Evidence:** Monte Carlo results show good finite-sample performance. In bilateral trade, geography and FTA channels are positive, the mixed channel is negative and time-varying, and indirect effects account for about 30%–40%.

Ongoing nonparametric extension

- Estimate the unknown coupling functions f_1 , f_2 , and f_3 using sieve bases rather than imposing a functional form.
- After removing fixed effects, the problem becomes a growing-dimensional IV/2SLS model with endogenous sieve spatial regressors.
- Under suitable growth and undersmoothing conditions, the finite-dimensional coefficients retain $\sqrt{N}$ -rate inference, while the estimated coupling functions converge at a nonparametric rate and admit pointwise inference.

Thank you

Questions and comments are welcome

Reference I

- Dargel, L. and Thomas-Agnan, C. (2023). Efficient estimation of spatial econometric interaction models for sparse od matrices.
- Jeong, H. and Lee, L.-f. (2024). Maximum likelihood estimation of a spatial autoregressive model for origin–destination flow variables. *Journal of Econometrics*, 242(1):105790.
- LeSage, J. P. and Pace, R. K. (2008). Spatial econometric modeling of origin-destination flows. *Journal of Regional Science*, 48:941–967.
- LeSage, J. P. and Thomas-Agnan, C. (2015). Interpreting spatial econometric origin-destination flow models. *Journal of Regional Science*, 55(2):188–208.
- Lin, Y. (2024). *Three Essays in Theoretical and Applied Spatial Econometrics*. PhD thesis, The Ohio State University.